

Analysis and Prediction on Data Science Salaries (2020-2025)

Kevin Garofalo¹, Giulio Lonati¹, Vincenzo Siano¹

Abstract

This project presents an analysis and predictive modeling study on global data science salaries from 2020 to 2025, using a publicly available dataset of job postings. The main objectives are to identify the main factors influencing salary levels and to develop accurate models for salary prediction. Exploratory data analysis revealed substantial differences across experience levels, job titles, company locations, and other employment attributes. To quantify these relationships, statistical analysis with Ordinary Least Squares regression was applied to assess the significance of different predictors. Building on these insights, the machine learning phase implemented and evaluated multiple regression approaches, including Linear Regression, Lasso Regression, Random Forest, XGBoost, LightGBM, CatBoost, and a stacked ensemble model. The analysis highlights the relative importance of features such as experience level, job title, company location and salary currency.

Keywords

Salaries — EDA — Regression — Gradient Boosting — Cross Validation — Feature Selection

¹CdLM in Data Science, Università degli Studi di Milano-Bicocca

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1. Introduction

In recent years, the demand for data science professionals has grown significantly across industries, reflecting the central role of data-driven decision making. Salary levels in this field vary widely depending on factors such as experience, job role, company size, and geographic location. The aim of this study is to identify the main drivers of salary differences in data science and IT roles. Additionally, we test some predictive models that can estimate salaries based on observable job and company attributes. The analysis relies on a publicly available dataset, which contains job postings and salary information for data science positions. The dataset covers the

period from 2020 to 2025 and includes variables such as job title, experience level, company size, location, and employment type. The dataset is updated daily and includes anonymized data both from surveys and job posting websites, gathering offers from countries all around the world.

The research begins with exploratory data analysis to highlight key patterns and differences between groups. Subsequently, both statistical modeling (OLS Linear Regression) and machine learning methods (OLS Linear Regression again, Lasso, Random Forest, XGBoost, CatBoost, and stacked ensembles) are employed to assess predictive performance.

The aim of this project is to provide both professionals and organizations in the data science field with a better understanding of global salary structures and the main drivers of remuneration, while also showing effective predictive modeling strategies for labor market analysis.

2. Dataset

This dataset captures real-world global salary data for jobs in Data Science field, including also Artificial Intelligence and Machine Learning positions, from 2020 to 2025 [1]. Each row represents a reported salary, enriched with attributes like job title, experience level, company size, remote work ratio, and geographic context. This data is valuable for analyzing salary trends over time, comparing roles across countries, and exploring compensation based on various factors.

Specifically, the dataset contains 151,445 observations and 11 columns. In fact, the variables taken into consideration are the following:

- **work_year:** the year the salary was reported. Covers salaries from 2020 through 2025.
- **experience_level:** the seniority level of the employee at the time of reporting.
- **employment_type:** the type of employment contract.

- **job_title:** the employee's specific job title (e.g., Data Scientist, ML Engineer, AI Engineer, Research Scientist).
- **salary:** the employee's gross annual salary in the original reported currency, before taxes and deductions.
- **salary_currency:** the currency in which the salary was originally paid (e.g., USD, EUR, GBP).
- **salary_in_usd:** the employee's salary converted into USD using 2025 exchange rates for standardized comparison.
- **employee_residence:** the country (ISO 3166-1 alpha-2 code) where the employee resides; this may differ from the company location, especially in remote roles.
- **remote_ratio:** indicates the percentage of remote work (i.e., 0, 50, 100).
- **company_location:** the country where the company or employer is headquartered.
- **company_size:** the size of the employing organization, which can be small, medium or large.

This dataset was compiled by aggregating salary data for Data Science, AI, and Machine Learning roles collected from multiple reputable sources, including the *Foorilla* website (ex AIJobs.net) salary survey [2], which provides anonymized, up-to-date compensation data from 2020 to 2025, and also industry-leading platforms and reports such as *365 Data Science*, *Payscale*, *KDnuggets*, and *Zip Recruiter*, to ensure comprehensive and diverse coverage. To create this unified dataset, the authors performed various operations. For instance, raw data from these sources was extracted and cleaned to remove duplicates and incomplete records, standardized by normalizing job titles and experience levels across datasets, converted all salaries into USD using the latest exchange rates for consistent comparison, categorized employment types,

company sizes, and remote work ratios for detailed analysis, and they validated for accuracy by cross-referencing multiple data points.

This provenance ensures the dataset's quality and usefulness for salary analysis, machine learning modeling, and career research worldwide.

3. Data Exploration

First of all, exploratory data analysis (EDA) and pre-processing steps were carried out to understand the dataset's characteristics, and prepare it for modeling, identifying potential issues, and, above all, uncover useful relationships for modeling. This section summarizes the main findings from the EDA part, grouped into three areas. Before diving into exploration, we needed to prepare the dataset for our analysis.

3.1 Pre-processing

We performed a small pre-processing task related to cleaning and modifying the values of some features to provide a clearer understanding and interpretability. The dataset does not present null values or typing errors, as it has been validated at the source. Therefore, our changes are summarized as follows:

1. we converted all country names from ISO2 to ISO3;
2. we extended all experience level, employment type, company size and remote ratio abbreviations to their full names.

3.2 Target Analysis

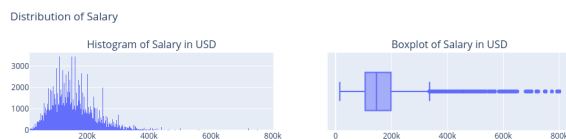


Figure 1. Distribution of the target variable (salary).

The target variable (**salary_in_USD**) was examined to assess its distribution and variability. An initial analysis of its distribution (Fig. 1) revealed

a significant positive skew, with a long tail of high salaries. This is a common characteristic of salary data, with most observations concentrated within a mid-range and a small number of extreme values (outliers).

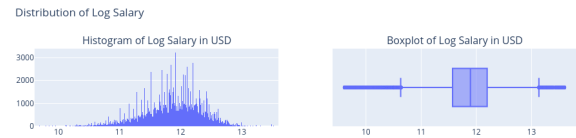


Figure 2. Distribution of log-transformed salary.

Practically, this means few people gain way more than the average. This pattern suggested the need for potential transformations or robust models to mitigate the influence of unusually high or low salaries. To normalize this distribution and better meet the assumptions of linear regression models, a logarithmic transformation was applied to the target feature. From Fig. 2, the resulting log-transformed salary distribution is more symmetrical and approximates a normal distribution, which is better for modeling, since many machine learning algorithms work better with normal response.

3.3 Feature Analysis

A univariate analysis was performed on the predictive features to understand their individual distributions, characteristics and potential impact on salaries. This analysis confirmed several imbalances in the dataset, as initially discussed.

In general, several features of the dataset show imbalances and potential biases, which may affect both the statistical analysis and the predictive models:

- **Work year:** records from 2024 and 2025 are overrepresented, likely reflecting the recency of data collection and/or the recent growth in data science roles. This imbalance may bias trend analyses by amplifying recent patterns and reducing the reliability of comparisons across years.
- **Employment type:** freelancer, part-time, and contractor positions are underrepresented compared to full-time ones. As a result, models

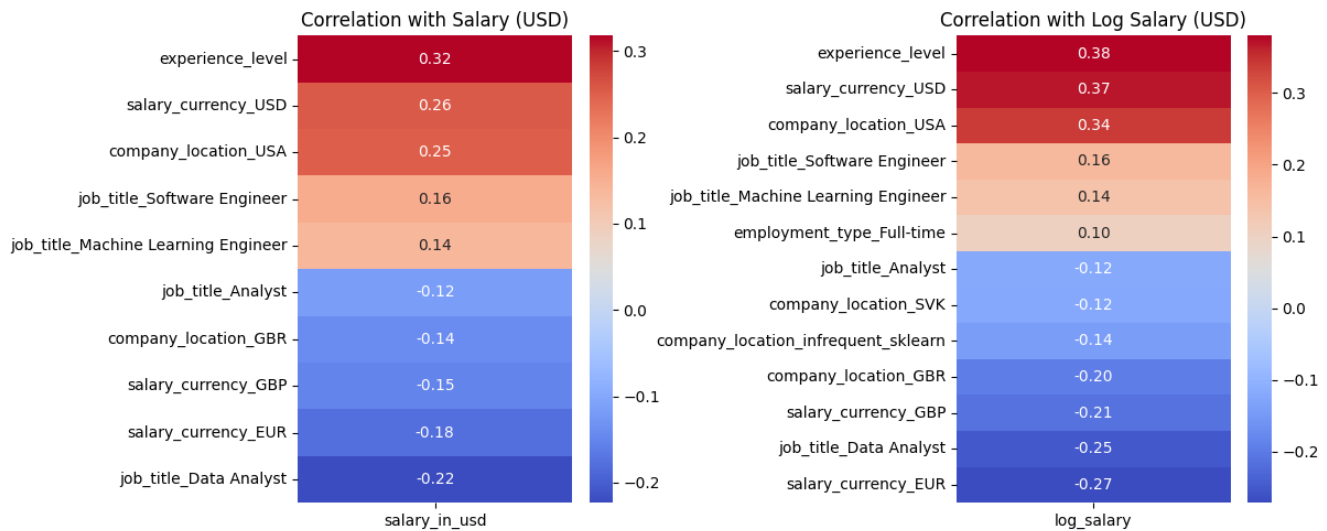


Figure 3. Correlation heatmap with the other features, both for salary in USD and log salary.

may generalize poorly for these categories and predictions could be less accurate.

- **Salary currency:** salaries in USD currency dominate the dataset, which may skew comparisons across currencies and reduce the generalizability to other countries.
- **Geographic distribution:** both employee residence and company location are heavily concentrated in the United States. This may lead to models that do not capture salary dynamics in less represented regions.
- **Remote ratio:** half-remote roles are pretty rare, and on-site positions are more common than fully remote ones. This could cause predictive models to underestimate the prevalence and salary impact of remote and hybrid roles.
- **Company size:** medium-sized companies represent the majority of entries, while small and large organizations are less frequent. Salary estimates for those companies may therefore be less reliable.
- **Salary anomalies:** reported salaries in certain countries appear unusually high, likely due to biases in the source websites, which are often specialized in data science and IT fields. Moreover, people who participated in the survey data were highly-skilled or working in companies above the average. These

outliers could distort the models' outcomes if not properly handled.

3.4 Multivariate Analysis

To explore the relationships between features and their impact on salary, several visualizations were created. Correlation analysis was carried out to identify associations between **encoded** categorical features and the target variable. Therefore, a correlation heatmap was created. As we can see from Fig. 3, **log salary** registered higher correlations with our features than **salary in usd**, suggesting the transformation can be helpful for modelling, especially when using linear models. Stronger positive relationships were observed between **salary** and **experience level**, **salary currency in USD**, and **company location USA**, while negative relationship with **salary currency EUR** and **job title Data Analyst**. These insights guided feature selection, informed expectations for regression modeling and, furthermore, helped identify any potential multicollinearity issues before modeling. In fact, we can also spot correlations between **salary currency** and **company location**: it makes sense to think the fact that a salary which has been paid in USD is correlated to the fact that salary comes from a company located in the USA, and so on. This led us to consider whether to drop one of these two variables to avoid collinearity.

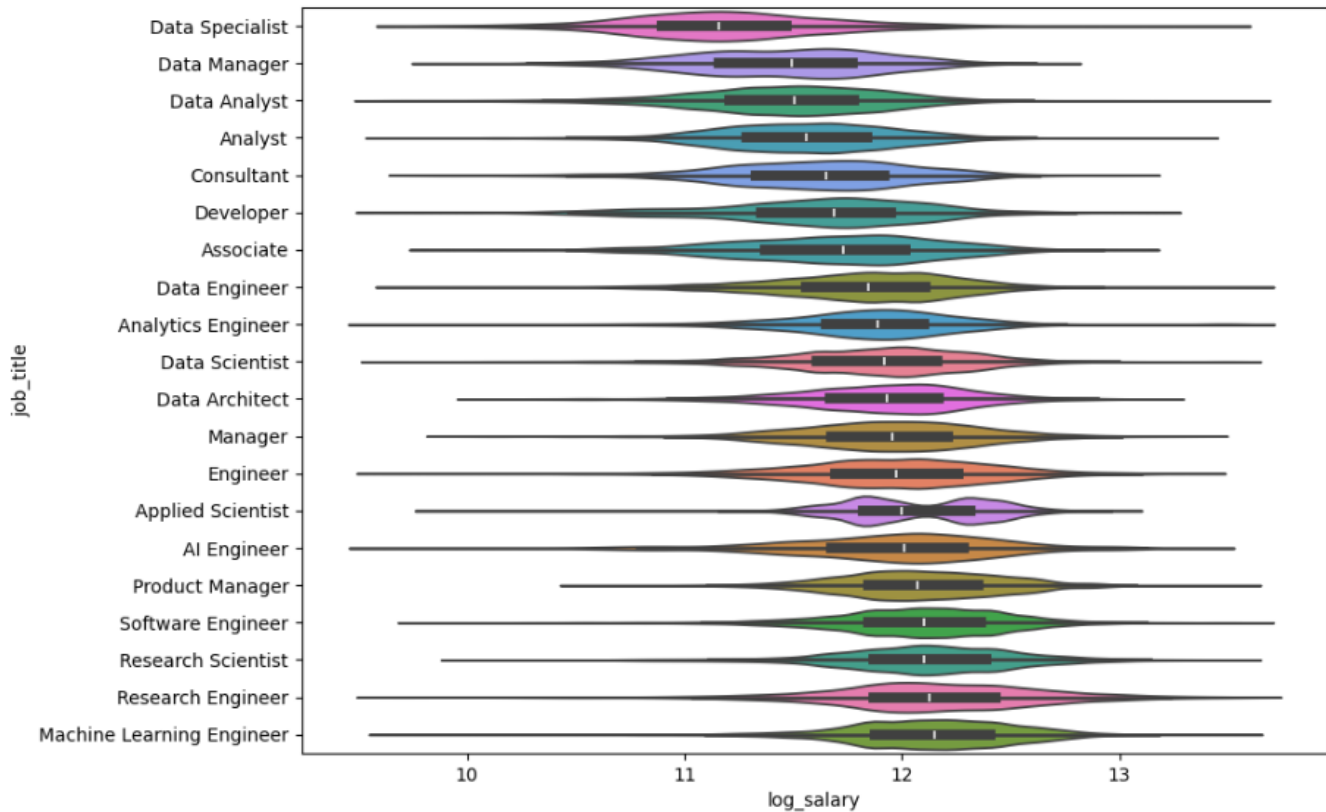


Figure 4. Violin plot of job title and log salary.

In addition to correlations, we also studied the interactions between our target and feature variables. First, we visualized these by using violin plots of **log salary** (after considering **salary in USD**) against the different categories of our features, limiting the visualized categories to the top-20 by frequency. The most interesting ones appeared to be **job title** (Fig. 4) and **experience level** (Fig. 5).

Job title The violin plots related to job title reveal noticeable differences in the median salary levels across roles. For instance, *Machine Learning Engineers*, on average, have the highest salary. However, at the same time, they show great variability within each job category, as indicated by the wide spread and long tails of the distributions. This makes reliable salary prediction more difficult. Job title is an important factor, but it is not a sufficient one.

Experience level As expected, there is difference

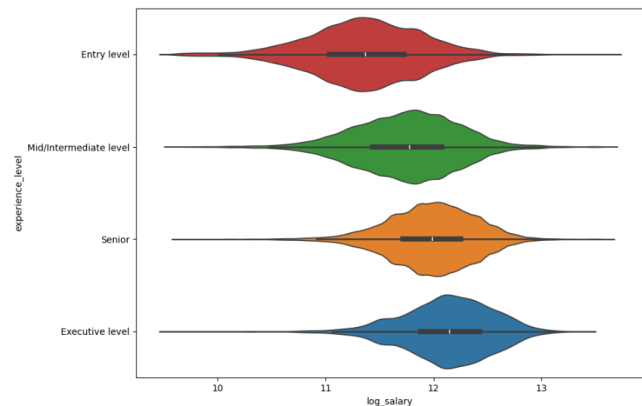


Figure 5. Violin plot of experience level and log salary.

between the salaries for the various categories of experience level. The shape of the distribution appears similar across all four categories considered, although salaries for entry-level positions are spread over a wider range.

Regarding the other features, we observe the following:

- **salary currency:** different currencies have different salary distributions; more in detail, salaries paid in *USD*, on average, are the highest, followed by salaries paid in other few currencies (e.g., *CHF* for Switzerland, and *SGD* for Singapore).
- **company location:** very similarly to the previous, different company locations have different salary distributions; more in detail, companies located in the *USA*, on average, pay the highest salaries.
- **employment type:** "Full-time" workers, on average, have higher salaries than "Contractor", which in turn have higher salaries than the other categories; in this case, grouping could take to loss of information.
- **remote ratio:** *Half-remote* roles show a higher variability and the median salary is lower than the other two; *On-site* and *Full-remote* jobs have similar median salaries and more concentrated distributions, though still with long tails; it doesn't look too much informative, considering the distributions of salary across these 3 categories are not well-separated.
- **company size:** similar to the previous.
- **work year:** on average, salaries from 2022 to 2025 are very similarly distributed and higher than salaries in 2020 and 2021, but still it doesn't look like a well-separated distribution.

Finally, we plotted other informative charts. First, referring to Fig. 6, we analyzed median **salary in USD** of each **job title** based on the **experience level** of the worker, which demonstrated that a combination of these two variables could be more effective in explaining salary variations than trying to do it with only one of them (though, this would take to have a huge number of encoded features). Similarly, we visualized a map of the median salary by **company location** around the world. Lastly, we observed the evolution of the proportion of **job title** throughout the years, noticing that we went from a very linear and basic naming of the jobs (e.g., *Data Scientist*, *Data Engineer*, *Data Analyst*), to a more granular and specialized one.

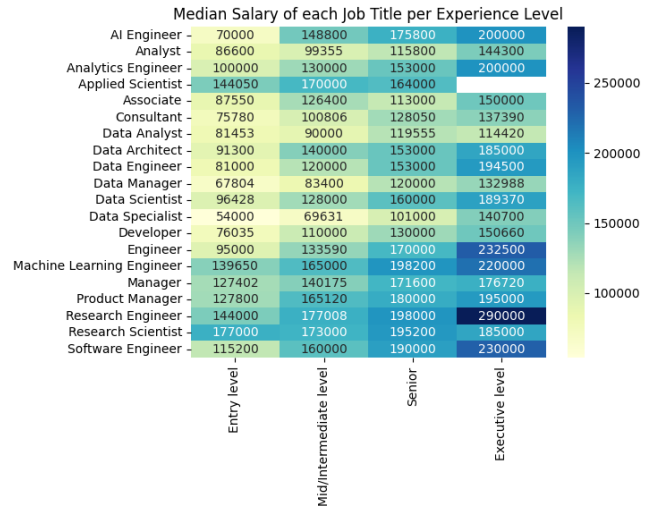


Figure 6. Median salary in USD of each job title per experience level of workers.

These analyses not only confirmed the importance of certain features but also revealed potential redundancies, later addressed in model selection.

4. Methodologies

This section outlines the analytical and predictive methodologies employed to investigate the factors influencing data science salaries and to build robust predictive models. We tried two different approaches:

1. A **statistical analysis** to interpret relationships between variables, aiming to build a simple and **interpretable** regression model.
2. A machine learning pipeline to accurately predict salary levels, aiming to build a more complex and **performant** regression model we could use for **prediction**.

4.1 Statistical Approach

To establish a baseline understanding of the relationship between predictors and salaries, we began with a statistical regression approach. Specifically, we used an **Ordinary Least Squares (OLS) Linear Regression** to assess how our features relate to salary outcomes. OLS is a linear regression method that aims to find the best-fitting line through the data by minimizing the sum of the squared differ-

ences between the observed and predicted values. The statistical analysis served two main purposes:

1. **Interpretability:** OLS regression allows us to quantify the marginal effect of each variable while holding others constant. Coefficients provide insights into how changes in categorical and numerical features affect salaries. OLS regression produces coefficients for each feature, which represent the expected change in the dependent variable (**salary in USD**) for a one-unit change in that feature, holding all other features constant. Furthermore, the statistical significance of each coefficient is tested (using **p-values**), allowing us to distinguish which factors have a genuine and non-random association with salary.
2. **Baseline:** it establishes a simple predictive model against which more complex ML models can be compared.

We checked regression assumptions (i.e., linearity, homoscedasticity, multicollinearity, and normality of residuals) to validate model suitability. To fit the model, we used the **statsmodels** Python package, which is well-designed for **inference**.

4.2 Machine Learning Models

For the predictive task, a variety of **Machine Learning models** were implemented to identify the most effective algorithm for this dataset. While **statistical regression** provides interpretable baselines, machine learning models are capable of capturing non-linear relationships and feature interactions that often arise in salary data.

Before training, we created a **scikit-learn** pipeline consisting of a preprocessing stage and a prediction model. In the preprocessing step, numerical features were standardized, categorical features were transformed using a *One Hot Encoder*, while ordinal features were handled with an *Ordinal Encoder*. This ensured that all input variables were in the correct format before being passed to the regression models. Furthermore, before training any model, we split our dataset into a **training set** and **test set**,

in order to have a set of fresh unseen data to test the performances of our final model. It is common to use 80% of the data for training and hold out 20% for testing. Each model was trained using the same train-test split to ensure comparability. We also employed a **5-fold cross validation approach**, which randomly splits the training set into 5 non-overlapping subsets called folds, then it trains and evaluates the model 5 times, picking a different fold for evaluation every time and using the other 4 folds for training. This technique allows us to evaluate our models' performance better and reduces the risk of overfitting [3].

4.2.1 Linear and Lasso Regression

Linear Regression After setting up the pipeline, we first applied a **Linear Regression** model, similar to the statistical regression performed earlier with **statsmodels**. The only difference was that, within the pipeline, we also applied a standardization to the numerical features in order to normalize their scale before modeling.

Lasso Regression Building on this baseline, we then implemented a **Lasso Regression**. Lasso (Least Absolute Shrinkage and Selection Operator) is a linear model that introduces an L1 regularization penalty, which helps prevent overfitting but can also drive some coefficients to zero, performing feature selection.

4.2.2 Random Forest

Random Forest is an ensemble learning method that constructs numerous decision trees on bootstrapped samples of the data and averages their predictions. This approach reduces variance compared to a single decision tree and generally improves robustness and predictive performance.

4.2.3 XGBoost

XGBoost (Extreme Gradient Boosting) is a highly efficient and popular implementation of the **Gradient Boosting** algorithm. Specifically, the latter is a powerful ensemble machine learning algorithm, also known as gradient tree boosting. Ensembles are constructed from decision tree models: trees

are added one at a time to the ensemble and fit to correct the prediction errors made by prior models. Models are fit using any arbitrary differentiable loss function and **gradient descent** optimization algorithm. This gives the technique its name, as the loss gradient is minimized as the model is fit, much like a **neural network**. There are many implementations of the gradient boosting algorithm available in Python. Perhaps the most used implementation is the version provided with the scikit-learn library. Additional third-party libraries are available that provide computationally efficient alternate implementations of the algorithm that often achieve better results in practice. Examples include the **XGBoost**, **LightGBM**, and **CatBoost** libraries. The main benefit of the XGBoost implementation is computational efficiency and often better model performance. Moreover, it includes built-in L1 and L2 regularization to prevent **overfitting** and is known for its **exceptional performance** and speed on tabular and similar structured datasets [4].

4.2.4 LightGBM

We also experimented with **LightGBM (Light Gradient Boosting Machine)** as part of our modeling phase. LightGBM is a library developed by Microsoft that provides an efficient implementation of the gradient boosting algorithm. It builds decision trees in a highly efficient way, using a leaf-wise growth strategy with depth constraints, and it is designed to be faster and more memory-efficient than traditional gradient boosting methods.

4.2.5 CatBoost

CatBoost (Category Gradient Boosting) is a library developed at Yandex that provides another efficient implementation of the gradient boosting algorithm. It employs an efficient method of encoding categorical data that helps to prevent target leakage and improves model performance. Given the number of categorical variables in this dataset, CatBoost is a particularly suitable candidate.

4.3 Evaluation Metrics

To assess model performance, we adopted two commonly used regression metrics:

- **MAE (Mean Absolute Error)**, which captures the average magnitude of the errors between predicted and actual salaries. Unlike **RMSE (Root Mean Squared Error)**, MAE is more robust to outliers, making it suitable for salary prediction where extreme salaries may skew results. MAE is calculated on the original salary scale (i.e., USD) to provide an intuitive and directly interpretable measure.
- **R²**, also known as the **coefficient of determination**, represents the proportion of the variance in the dependent variable that is predictable from the independent variables. It provides a measure of how well the model explains the variability of the data, with values ranging from 0 to 1. Higher values indicate better fit. This metric was calculated on the **log-transformed** salary in USD target variable.

Note that R² has been considered just for comparing with the baseline Linear Regression applied in the statistical analysis. These metrics provide complementary perspectives: R² evaluates explanatory power, while MAE quantifies predictive accuracy in absolute terms.

4.4 Fine-Tuning and Stacking

We decided to tune the parameters of the **LightGBM** model because of its speed and efficiency. One option would be to fiddle with the hyperparameters manually, until finding a great combination of hyperparameter values. Though, this would be a very tedious work. For this reason, it's preferable to use scikit-learn's **RandomizedSearch**. It evaluates a fixed number of combinations, selecting a random value for each hyperparameter at every iteration. Thus, we tuned the following key **LightGBM's parameters**:

- number of trees in the model; more trees can improve performance but increase training time. Lower values make learning slower but can improve accuracy.
- learning rate, which controls the step size at each iteration while updating weights.

- maximum depth of each tree, remembering that a deeper tree can capture more complex patterns but may overfit.
- number of leaves per tree, controlling model complexity; more leaves allow more complex splits, affecting both accuracy and overfitting risk.
- minimum number of data points required in a leaf node; larger values prevent overfitting by making leaves more general.

Next, we implemented a **stacking ensemble** to combine the strengths of different algorithms to create a more powerful final meta-model. Precisely, this involves training multiple base models (i.e., Random Forest, XGBoost, LightGBM and CatBoost), using a meta-learner to aggregate their predictions. This approach exploits the diversity of the individual models to achieve improved predictive performance when compared to any single model alone.

5. Results

5.1 Statistical Analysis

The initial regression analysis using OLS Regression provided a baseline understanding of how the selected predictors influence salaries.

Table 1. Models comparison.

#	Target	R^2	Adjusted R^2	DF
1	Salary	0.298	0.295	546
2	Log Salary	0.426	0.424	546
Final	Log Salary	0.341	0.341	14

As we can see in Table 1, we started using the **salary in USD** as target (all the other features as predictors), obtaining a poor ($R^2 \approx 30\%$) and not interpretable (degrees of freedom = d.f. ≈ 500) model.

After, we reached an intermediate version using the **log salary** as target instead, obtaining a huge jump in explained variance ($R^2 \approx 43\%$), but still too many variables (d.f. ≈ 500) to interpret the model; eventually, we dropped irrelevant features and grouped under-represented categories, reaching a moderately performing ($R^2 \approx 34\%$), but finally interpretable (d.f. = 14) model.

The final model, selected for better interpretability, is characterized by the experience level (encoded ordinal), reduced versions of job title (with top-10 most frequent one hot encoded categories) and company location (with top-4 most frequent categories, always one hot encoded). Following, we report the most interesting insights:

- The model output has **high statistical significance**: p-values of the F-test and of the t-test of all variables are lower than all common significance levels (e.g., 0.05, and 0.01).
- **R^2** and **Adjusted R^2** are both 0.341, meaning our features are able to explain 34.1% of the total variance of the target **log salary**. This is an acceptable score in a field like **salaries analysis**, considering the frequent presence of **noise**, which our model is not able to capture.
- Residuals are close to normal (**Skew** = 0.020, and **Kurtosis** = 3.811), meaning the model respects the initial linear regression assumptions.

Since the target is log-transformed, let's interpret **coefficients** as percentage changes:

1. **Job titles** with the most positive effect on salary with respect to the **baseline** job title (i.e., **Data Analyst**), while keeping all other predictors fixed, are as follows:
 - *Research Scientists*, on average, earn 67% more.
 - *Machine Learning Engineers*, on average, earn 62% more.
 - *Product Managers*, on average, earn 56% more.
 - *Software Engineers*, on average, earn 53% more than a Data Analyst.
2. **Company locations** with the highest impact on salary with respect to baseline company location (i.e., **Australia**), while keeping all other predictors fixed, are as follows:
 - **American** companies' salaries are, on average, 7.8% higher.

- Salaries of companies that are not in the USA, Canada, United Kingdom or Australia, on average, are 51.1% lower.
 - **UK** companies' salaries are, on average, 35.6% lower.
 - **Canadian** companies' salaries are, on average, 11.6% lower than Australian ones.
3. Similarly, each increase of **experience level** (i.e., Entry level → Mid/Intermediate level → Senior → Executive level) is associated with a 22.7% increase in salary.

In general, the R^2 score indicated that the linear model could explain only a modest proportion of the variance in salaries, suggesting that the relationship between predictors and salary is **not strictly linear**. This highlighted the need for more flexible models capable of capturing **non-linear** interactions and complex patterns in the data.

5.2 ML Models Comparison

A set of machine learning models was evaluated using the same cross-validation configuration, with results in terms of MAE (reported in the original dollar scale for interpretability), and R-squared (related to the log-transformed salary).

Table 2. Comparison of the Machine Learning models using cross-validation.

Model	MAE (\$)	R^2
Linear Regression	44,680.349	0.417
Lasso Regression	44,998.109	0.405
Random Forest	44,357.172	0.433
XGBoost	44,139.255	0.438
LightGBM	44,513.890	0.417
CatBoost	44,201.583	0.435
Fine-Tuned LightGBM	44,594.290	0.414
Stacking	44,056.901	0.441

The results are summarized in Table 2, and described below:

- **Linear Regression:** Using all variables, we obtained a **Mean Absolute Error (MAE)** of \$44,680. This indicates that the model's

prediction deviates, on average, by around \$44k from the real salary.

- **Lasso Regression:** Achieved a **MAE** of \$44,998.109, selecting 104 features out of 533. This shows that Lasso can reduce model complexity by eliminating less informative predictors, though the predictive performance is slightly lower than standard Linear Regression.
- **Random Forest:** Reached a **MAE** of \$44,357.172. While this is a modest improvement, the model is by far slower to train compared to linear models due to the ensemble of many trees.
- **XGBoost:** Provided a **MAE** of \$44,139.255, outperforming Random Forest due to its iterative boosting approach.
- **LightGBM:** Achieved a **MAE** of \$44,513.890, performing slightly lower than XGBoost, but offering much faster training and inference times.
- **CatBoost:** Obtained a **MAE** of \$44,201.583, comparable to Random Forest and slightly below XGBoost.
- **Fine-Tuned LightGBM:** got worse **MAE** than the original, with \$44,594.290, suggesting hyperparameters were not the main issue.
- **Stacking Model:** Combining some of the above models resulted in a **MAE** of **44,056.901**, representing the best predictive performance by leveraging the strengths of multiple algorithms.

The best model in terms of performance happened to be the **Stacking Model**; the best model in terms of speed at training and inference times happened to be **LightGBM**, with slightly lower performance than the latter. For the final evaluation, the **Stacking Model** was retrained on the entire train set and assessed on the held-out test set, providing slightly higher values than the validation metrics (**MAE** = **43,952.808**, and **R^2** = **0.450**). Overall, despite increasing model complexity, using advanced ensemble methods, performing fine-tuning and model stacking, it remains challenging to achieve **MAE** values below \$44,000 and **$R^2$** values above **45%**.

This suggests that a substantial portion of salary variability is influenced by factors that are either not captured in the dataset or inherently difficult to model.

5.3 Feature Importance

To better understand model decisions, feature importance was extracted from a pair of ensemble models, namely, **Random Forest** and **XGBoost**.



Figure 7. Random Forest’s Feature Importance, similar to the statistical approach’s insights.

Across these models, looking at Fig. 7, some features emerged as important predictors of salary with tiny variations:

- **Experience level:** unsurprisingly, the importance of experience level confirms that **seniority** and career progression are the primary determinants of compensation in the data science field.
- **Geographic and economic indicators:** the high importance of two closely related features, company location in the **United States** and salary currency in **USD**, indicates a substantial **high salary** for roles based in the United States, where salaries are denominated in USD. These features serve as strong proxies for the high-paying North American market.
- **Job title:** specific jobs, particularly those related to core roles, like *Data Analyst*, *Machine Learning Engineer*, *Engineering Manager*, *Research Scientist*, and *Data Specialist*, also contributed significantly to the models’

predictions, highlighting the specialization’s impact on compensation.

These findings from the machine learning models align with and reinforce the insights from the statistical analysis. Interpreting these features highlights expected labor market trends, as experience and job title are driving higher compensation. They provide a robust and consistent picture where the primary drivers of data science salaries are an individual’s experience level and the geographic location of the specific role, with a significant advantage for those employed in the United States. The consistency of feature importance across Random Forest and XGBoost strengthens confidence in these insights, highlighting a clear group of primary salary drivers.

6. Conclusions

This study examined global data science salaries through two complementary perspectives: a statistical approach and a set of machine learning models. From the statistical perspective, both visualizations and linear regression provided a baseline for understanding salary variation: the main results demonstrated that the linear model explained only a modest share of the variability, with R^2 values in the range of 0.29-0.42, with the final model obtaining a score of 0.34. This limited explanatory power underscores the complexity of salary determination, which is likely influenced by additional unavailable, and unobserved factors such as negotiation strategies, company culture, or broader labor market conditions. From the machine learning perspective, evaluation was centered on predictive accuracy as measured by Mean Absolute Error (MAE). Tree-based ensemble models, such as Random Forest, XGBoost, CatBoost, and especially Stacking, consistently reduced prediction error compared to linear methods. However, while these models improved accuracy, the magnitude of the gains remained moderate, indicating that even advanced algorithms face constraints and limits when predicting salary outcomes from the available features alone.

Despite these limitations, the analysis provides useful insights. Feature importance analysis high-

OLS Regression Results							
Dep. Variable:	log_salary	R-squared:	0.341				
Model:	OLS	Adj. R-squared:	0.341				
Method:	Least Squares	F-statistic:	5595.				
Date:	Sat, 13 Sep 2025	Prob (F-statistic):	0.00				
Time:	14:58:31	Log-Likelihood:	-74272.				
No. Observations:	151445	AIC:	1.486e+05				
Df Residuals:	151430	BIC:	1.487e+05				
Df Model:	14						
Covariance Type:	nonrobust						
		coef	std err	t	P> t	[0.025	0.975]
	Intercept	11.2572	0.016	685.404	0.000	11.225	11.289
	C(job_title_reduced)[T.Data Engineer]	0.2081	0.004	48.608	0.000	0.200	0.217
	C(job_title_reduced)[T.Data Scientist]	0.2595	0.004	62.792	0.000	0.251	0.268
	C(job_title_reduced)[T.Engineer]	0.3130	0.005	65.293	0.000	0.304	0.322
	C(job_title_reduced)[T.Machine Learning Engineer]	0.4848	0.005	94.175	0.000	0.475	0.495
	C(job_title_reduced)[T.Manager]	0.3370	0.005	63.142	0.000	0.326	0.347
	C(job_title_reduced)[T.Other]	0.2155	0.003	62.174	0.000	0.209	0.222
	C(job_title_reduced)[T.Product Manager]	0.4448	0.008	53.381	0.000	0.428	0.461
	C(job_title_reduced)[T.Research Scientist]	0.5125	0.007	70.040	0.000	0.498	0.527
	C(job_title_reduced)[T.Software Engineer]	0.4281	0.004	100.571	0.000	0.420	0.436
	C(company_location_reduced)[T.CAN]	-0.1228	0.017	-7.234	0.000	-0.156	-0.090
	C(company_location_reduced)[T.GBR]	-0.4394	0.017	-25.374	0.000	-0.473	-0.405
	C(company_location_reduced)[T.Other]	-0.7143	0.017	-41.628	0.000	-0.748	-0.681
	C(company_location_reduced)[T.USA]	0.0748	0.016	4.603	0.000	0.043	0.107
	enc_experience_level	0.2052	0.002	135.230	0.000	0.202	0.208
Omnibus:	2177.449	Durbin-Watson:	1.912				
Prob(Omnibus):	0.000	Jarque-Bera (JB):	4158.996				
Skew:	0.021	Prob(JB):	0.00				
Kurtosis:	3.811	Cond. No.	77.5				

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 9. Summary table of the statistical interpretable model, with just 14 degrees of freedom and a moderate proportion of explained variance.

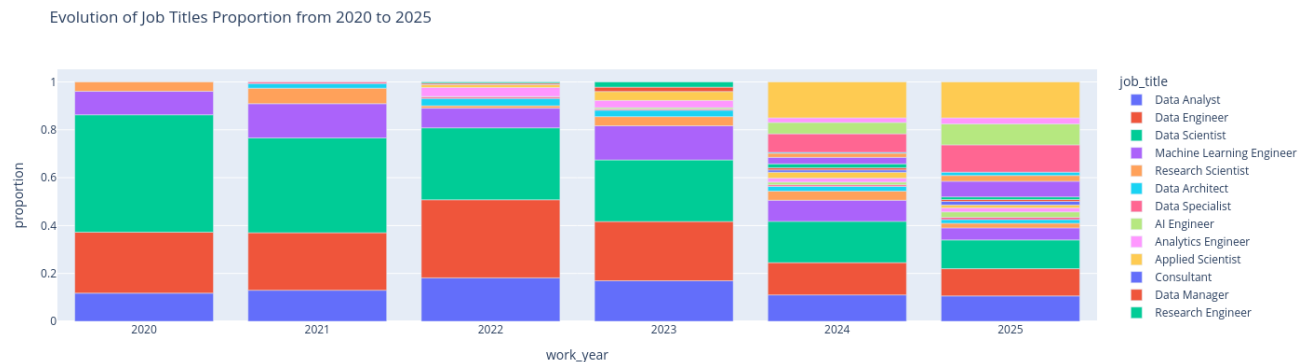


Figure 10. Evolution of the proportion of different job titles from 2020 to 2025: we can see how we went from a very simple and general division (a lot of data scientists, data analysts and data engineers) to a more specialized and precise one.

Median Salary by Company Location



Figure 11. Map of the median salaries for top-30 most frequent countries in our dataset. As expected, American companies provide the highest ones.

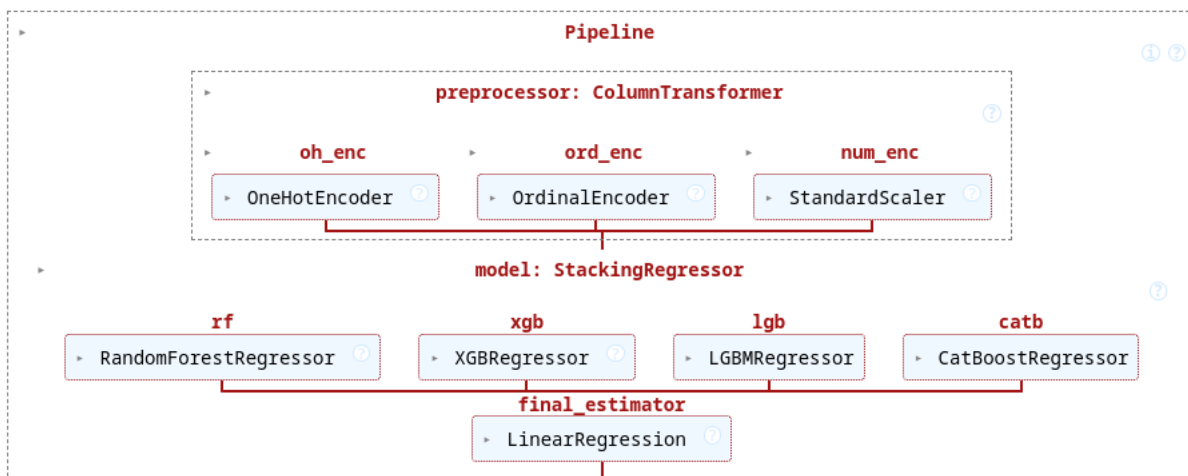


Figure 12. Pipeline of stacking model, our best performing regressor.

References

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